

# Shop Assistant

## Help & Reference Guide

The Shop Assistant application is a collection of calculations, design, and layout tools for use by the novice woodworker, home owner, and do-it-yourselfer. The various tools available are covered in this guide with photos, sketches, and explanations.

## How This App Works

- Most calculators support multiple unit systems (decimal inches, fractional inches, millimeters, feet, etc.) via a unit toggle near the top of the card.
- Many layout calculators (Picket Layout, Spindle Layout, Wall Framing, Diagonal Slats) show a live diagram, and let you select an individual piece from a dropdown to see its exact position, length, and cut angles.
- Cards with a “Saved Layouts” section let you save your current inputs under a name and reload them later.
- Cards with a “Print / Save as PDF” button generate a printable summary of that card's inputs and results, including its diagram where applicable.
- Use “Data Backup” at the top of the page (Export All Data / Import Data) to move your saved layouts and settings between devices.

## 01 — Saw Cuts

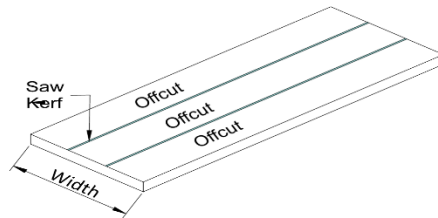
Setup calculations for common table saw and miter saw operations.

### Board Split

Rip a board into equal-width offcuts, accounting for blade kerf.

Usage:

- Select desired units
- Enter Board Width
- Choose number of desired offcuts
- Choose blade
- Blade library can be edited to add/delete blades
- Offcut width and number of cuts are displayed

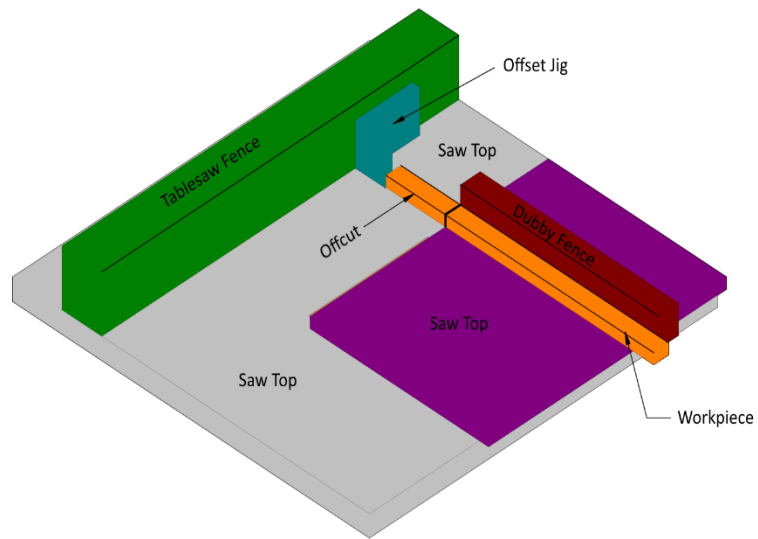


### Dubby Sled Offcut

Find the fence setting for cutting to length with Dubby sled

Usage:

- Select desired units
- Enter desired length
- Select the jig
- Jig library can be edited to add/delete blades
- Fence setting is displayed



## Dado Cut

Plan dado width/depth using multiple passes of a single blade.

Usage:

- Select desired units
- Enter dado stack width
- Enter desired dado width
- Select the jig
- Jig library can be edited to add/delete blades
- Fence setting is displayed

## Unequal Board Miter

Miter angles for joining two boards of different widths.

Usage:

- Select desired units
- Enter board A width
- Enter board B width
- Enter corner angle
- Select the jig
- Jig library can be edited to add/delete blades
- Fence setting is displayed

## Crown Moulding

Spring angle and blade tilt/miter settings for crown molding. Cuts assume that molding is lying flat on the miter saw table.

Usage:

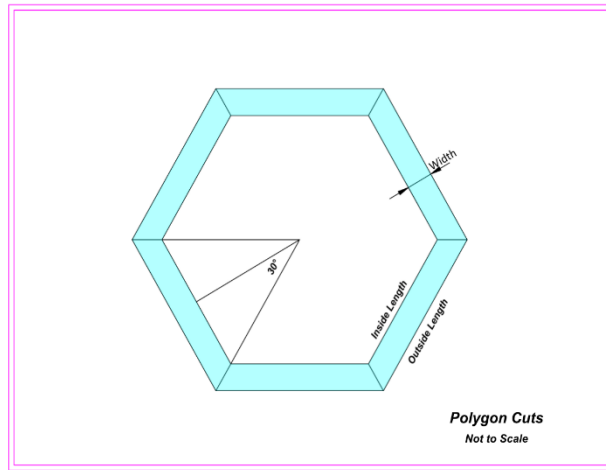
- Select desired units
- Enter spring angle. If spring angle isn't known, use the spring angle calculator
- Enter corner angle
- Select inside or outside corner
- Miter angle and bevel angle are displayed

## Polygon Cuts

Miter angle, side length, and radius for any regular polygon.

Usage:

- Select desired units
- Enter number of side desired
- Enter board width
- Enter desired side length
- Choose whether side length is inside or outside of the polygon
- Displays the following:
  - Miter angle
  - Interior corner angle
  - Outer side length (long point)
  - Inner side length (short point)
  - Radius to outer long point



## Diagonal Slats

Per-slat blank lengths and miter angles for diagonal slat patterns on a panel, including corner and edge pieces. Slat spacing can be specified.

Usage:

- Select desired units
- Enter panel width
- Enter panel height
- Enter slat angle
- Enter slat width
- Enter gap between slats
- Enter slat direction. Choices are standard and reversed. For standard direction, the slats rise from left to right. For reverse, the slats fall from left to right.
- Enter layout choice of single panel or split in half. If split in half is selected, the panel width is divided and the right half mirrors the left half.
- Displays the following:
  - Number of slats
  - Longest blank length
  - Total length of slats

- Combo box that displays individual slats by number. When a slat is selected, its length is displayed. In addition, the miter angles for both the left and right ends of the slat are displayed along with the attachment (bottom, top, side) and miter angle for each end. When a slat is selected its location in the image is highlighted.

## 02 — Tool Adjustment

Calibration and fine-tuning calculators for shop machinery.

### Tablesaw Fence

Fence offset calculations for angled/tapered cuts.

## 5-Cut Adjustment

Classic 5-cut method for squaring a miter/crosscut sled.

## Planer Adjustment

Planer depth-of-cut and material removal tracking.

## Router Shims

Shim combinations to fine-tune a router bit or fence to an exact height/offset.

## Clamps & Shims

Reference calculator for clamping pressure and shim stacking.

# 03 — Design

Layout and proportion calculators for furniture and structural design.

## Picture Framing

Mat widths and reveal for multi-mat picture frames.

## Drawer Design

Drawer box dimensions accounting for slide clearance and material thickness.

## Rafter Design

Rafter length, seat cut, and ridge/tail cuts for a given pitch and span.

## Stairs & Steps



Rise/run, stringer layout, and code-check for stair design.

### Crossbuck Design

Layout for X-brace (crossbuck) door and gate patterns.

## 04 — Layouts

Spacing and positioning calculators for repeating elements.

### Object Spacing

Even spacing of objects across a given length.

### Spindle Layout

Baluster/spindle spacing between two rails, including corner spindles.

### Picket Layout

Picket spacing for rectangular fence sections and sunburst gate patterns.

### Square-Up (3-4-5 Method)

Verify a corner is square using the 3-4-5 triangle method, with tape-measure running totals.

### Wall Framing

Stud layout for straight and raked (gable-end) walls, including corner framing.

### Wheel Segments

Segment angles and lengths for a segmented wheel/circle glue-up.

### Circular Pattern

Even angular spacing of items around a circle.

### Rectangular Pattern

Grid spacing of items across a rectangular area.

### Interval Layout

General repeating-interval spacing calculator.

## 05 — Quantities

Material takeoff calculators for estimating job quantities.

### Area Shape

Area and perimeter for common shapes.

### Wall Shape

Wall area accounting for door/window openings.

### Studs

Stud count for a given wall length and spacing.

## Joists

Joist count and span for floor/ceiling framing.

## Decking

Decking board quantity and layout for a deck surface.

## Roofing

Roofing material quantities based on roof area and pitch.

## Suspended Ceiling

Grid and tile quantities for a drop ceiling.

## Brick & Block

Brick/block count for a wall of given dimensions.

## Drape Panels

Fabric panel quantities for window treatments.

# 06 — Calculators

General-purpose reference calculators used across many projects.

## Unit Convert

Live conversion between decimal inches, fractional inches, millimeters, feet, meters, gallons/quarts/pints, and pints/ounces/drops.

## Object Clearance

Required clearance between objects.

## Shelf Sag

Predicted shelf deflection under load for a given material and span.

## Cabinet Stability

Tip-over risk estimate for a cabinet based on weight and dimensions.

## Golden Ratio

Golden-ratio proportion calculator for aesthetic layout.

## Date Math

Date arithmetic (adding/subtracting days, finding intervals).

## Complementary Angle

Complementary and supplementary angle lookup.

## Slopes

General slope/angle reference calculator.

## Wrench Selection

Wrench size lookup for common fastener sizes.

## Slope / Grade

Convert between slope ratio, degrees, and percent grade.

## Stair Rise & Run

Quick rise/run calculation for stairs.

## Feet-Inches n Decimal

Convert between feet-inches and decimal feet/inches.

## Spirit Levels

Bubble-level accuracy/reading reference.

## Survey Levels

Elevation calculations for a surveying level, in decimal feet, feet-inch, or millimeters.

## Wire Sizing

Wire gauge selection based on amperage, distance, and voltage drop.

## Curve Solver

General curve/arc geometry solver.

# 07 — Data & Information

Reference tables for common hardware and building specifications.

## Wood Screws

Wood screw size and pilot-hole reference.

## Tap Drill

Tap drill size reference for cutting internal threads.

## Kreg Screws

Kreg pocket-hole screw selection by material thickness.

## Lag Bolts

Lag bolt sizing and pilot-hole reference.

## Fasteners

General fastener reference table.

## Drill Bits

Drill bit size reference and lookup.

## Furniture Dimensions

Standard/ergonomic furniture dimension reference.

## Span Tables

Structural span table reference for joists/rafters.

# 08 — Paint & Chemical Mixes

Mixing and dilution calculators for finishes and chemicals.

## Shellac

Shellac cut ratios and dilution, including a Flakes/Dilute mode.

## Latex

Latex paint mixing/dilution reference.

## Wood Stains

Saved database of wood stain colors and brands.

## Stain Modification

Adjusting an existing stain mix toward a target result.

## Chemical Mixes

Saved database of chemical dilution ratios (e.g. cleaners).

## Color Mixing

Suggests up to 3 base colors (from Red, Orange, Yellow, Green, Blue, Indigo, Violet, Purple, White, Black) and mixing percentages/ounces to approximate a target hex color. This is a simplified linear-blend approximation, not a true pigment simulation.

App-wide settings and this help reference.

### **Colors**

Change the app's color theme. Saved automatically and applies everywhere.

### **Help**

This document.